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Efficacy of photobiomodulation for the prevention and treatment of chemotherapy-induced oral mucositis in pediatric patients with hematologic cancers: a randomized clinical trial

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Abstract

Background Oral mucositis (OM) is a debilitating treatment-related complication that affects most children undergoing chemotherapy. This dual-arm clinical trial aimed to evaluate the efficacy of photobiomodulation (PBM) for prevention and photodynamic therapy (PDT) using indocyanine green (ICG) and low-level laser for treating methotrexate (MTX)-induced OM in pediatric patients with hematologic cancers.

Methods In the prevention arm, patients were initially assigned to a control group (received no laser) and subsequently reassigned to a PBM group in their next cycle, allowing intra-patient comparison under similar chemotherapy protocols. In the treatment arm, patients diagnosed with OM were randomly assigned to either a control group (sham laser) or a PDT group.

Results In the prevention arm, PBM led to a significant reduction in the incidence of OM from 66.67% to 6.67% ($p < 0.05$). In the treatment arm, the PDT group showed a significant reduction in the scores of all scales (WHO, NCI, and WCCNR) at both 3 days and 5 days after starting the treatment compared to the control group ($p < 0.05$). Moreover, the reduction in the NCI score from the third to the fifth days after the starting treatment is higher in the PDT group compared to the control group ($p < 0.05$).

Conclusion PBM appears to be a safe and effective approach for the prevention of MTX-induced OM in pediatric patients with hematologic cancers. Additionally, PDT using ICG and a low-level laser demonstrates promising results in treating OM lesions. Therefore, these findings highlight the efficacy of PBM for managing MTX-induced OM in this patient group.

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Trial registration The study has been registered on irct.behdasht.gov.ir on 4 December 2022 (Trial Registration Number: IRCT20120101008585N16).

Keywords Hematology, Indocyanine Green, Laser Therapy, Mucositis, Neoplasms, Pediatrics, Photobiomodulation, Photodynamic Therapy

Introduction

Oral mucositis (OM) is the most important side effects of oncologic therapy [1]. It originates in the submucosal layer and presents with clinical erosions and/or ulcerations, resulting in varying degrees of pain. The severity of the condition can lead to disability and impair the patient's quality of life. In severe cases, OM may cause inadequate food intake, leading to substantial nutritional deficiencies and weight loss, potentially necessitating parenteral nutrition. Chemotherapy (CT)-induced OM can also be associated with super infections, further increasing the risk of bacteremia and sepsis. Furthermore, the discontinuation of anti-cancer therapy in some patients due to OM can affect their overall survival chance. Therefore, prevention and treatment of OM are essential for the clinical management of cancer [2].

Approximately 50 percent of cancer patients receiving CT medications such as 5-fluorouracil (5-FU), busulfan, cisplatin, doxorubicin, etoposide, melphalan, methotrexate (MTX), taxanes and vinblastine develop oral complications [3]. On the other hand, the incidence and severity of CT-induced OM vary considerably, with younger children being at high risk. Moreover, pediatric patients with hematologic tumors are more prone to developing severe OM compared to those with solid tumors [2]. MTX is a specific antimetabolite at the cell cycle with a significant cytotoxic effect on the oral mucosa, contributing to OM development. Despite its mucotoxicity, MTX has been demonstrated to be an effective drug in the treatment of various childhood malignancies, including acute lymphoblastic leukemia (ALL), lymphomas and osteosarcoma (OS) [3, 4]. Although several options exist for the prevention and treatment of OM, there is no clear consensus or any agreement regarding the best option, particularly in young patients.

Photobiomodulation (PBM) refers to changes in cellular activity and transformation induced by radiation under certain conditions [5]. Targeted application of low-level lasers or light-emitting diodes to the oral mucosa is a non-invasive method that induces analgesic, anti-inflammatory, and wound-healing effects in the tissue by modulating reactive oxygen species and activating transcription factors [6]. The recent update of the Multinational Association of Supportive Care in Cancer and International Society of Oral Oncology (MASCC/ISOO) clinical practice guidelines for the management of mucositis in 2020 recommended the PBM using low-level laser therapy (LLLT) (applied to the oral mucosa) for OM prevention in adult patients undergoing hematopoietic stem cell transplantation (HSCT) conditioned with high-dose CT [7]. However, the lack of randomized clinical trials and the significant variability in PBM parameters prevent the development of a guideline for pediatric patients [3].

Recently, interest in photodynamic therapy (PDT) in oral medicine has advanced. In more detail, the photosensitizer dyes can absorb light at specific wavelengths, inducing an excited reaction that promotes tissue healing. Therefore, in PDT, a photosensitizer dye is first applied to the target tissue, followed by exposure to a laser beam of the corresponding wavelength. Its effectiveness has been demonstrated as a promising approach in a few animal and human studies [8, 9]. Notably, the choice of photosensitizer is a critical factor in PDT success. Different types of photosensitizers, each with a specific excitation wavelength, have been used in oral diseases. Indocyanine green (ICG), a fluorescent dye widely used for diagnostic purposes, has an optimal peak absorption at 810 nm, which is near the absorption peak of most diode lasers. This property makes ICG a promising photosensitizer for PDT [10].

Given the need for approaches to manage CT-induced OM in pediatric patients, this dual-arm clinical trial aimed to evaluate the efficacy of PBM using a novel clinical protocol for the prevention and PDT using ICG and a low-level laser for the treatment of MTX-induced OM in pediatric patients with hematologic cancers.

Methods and materials

A total of 45 subjects were enrolled in this dual-arm clinical trial, which employed a crossover design in the prevention arm and a parallel design in the treatment arm. The sample size estimation was based on a pilot study involving 5 participants per group, using a significance level of 5% ($\alpha = 0.05$) and a statistical power of 80% ($1\text{-}\beta = 0.8$). The results of the pilot study in the treatment arm indicated that a minimum of 15 participants per group was required, based on the comparison of mean pain score reduction (according to the World Health Organization (WHO) scale) from day 1 to day 5 between the PDT group (-1.58 ± 0.5) and the control group (-0.76 ± 0.85). Although the prevention arm of the study required

a smaller sample size ($n = 8$ per group), we decided to include 15 participants in all four groups for consistency across the study arms. They were selected from patients admitted to the Amir Oncology Hospital from January 2023 till September 2024. The study followed CONSORT guidelines and was conducted with the approval of the Research Ethics Committees of the School of Dentistry, Shiraz University of Medical Sciences (IR.SUMS.DENTAL.REC.1401.486). The study was registered on irct.behdasht.gov.ir (IRCT Id: IRCT20120101008585 N16). The inclusion criteria for the study were as follows: 1. Subjects (both males and females up to 18 years old) who had leukemia; 2. Subjects were scheduled to undergo repeated CT cycles with similar drugs and dosage including high-dose MTX. The exclusion criteria were any medical conditions that could impair wound healing, uncontrolled comorbidity (kidney, liver, pulmonary, or heart diseases), photosensitivity, epilepsy, allergy to ICG, and trismus. Moreover, participants who had previously received prophylactic LLLT were excluded from all groups in this dual-arm study.

Eligible subjects were scheduled to participate in the study after obtaining informed consent from their legal guardians. Based on the aims of the present study, each subject was initially allocated into four groups for the prevention and treatment of OM by an academic member who was not involved in this research as follows:

Prevention of OM

Before any intervention, patients were initially allocated to a control group which was not received PBM. Subsequently, we changed their group allocation to a PBM group in their next CT cycle, which followed a similar protocol, in order to minimize the influence of patient related risk factors—such as genetic predisposition, oral hygiene, and body mass index—on the incidence of OM. Therefore, no formal randomization was applied due to the crossover design. This crossover design allowed each subject to be allocated to both groups over time, enabling the evaluation of the PBM efficacy for the prevention of OM in the same patient under similar CT protocols. In the PBM group, LLLT was administered using a diode laser (Klas-DX—Intelligent LLLT Dose Unit, Konf, Taiwan) with a continuous wave length of 808 nm, a spot size of 0.5 cm^2, an energy density of 20 J/cm^2, an accumulated energy of 220 J per session, and an irradiation time of 40 s per point (power: 250 mW). The protocol was applied prophylactically on days 1 and 3, starting from the initiation of CT. In more detail, they were irradiated with a low-level laser at five anatomical sites in the oral cavity including the labial mucosa (2 points on the upper lips and 2 points on the lower lips), buccal mucosa (5 points on each side), tongue (4 points on the ventral and lateral sides), soft palate (1 point on each side), and floor of the mouth (1 point on each side). The device was used following the manufacturer's instructions. During the sessions, the patients and operator also used protective eyeglasses.

It is important to note that all subjects received oral hygiene instruction prior to the initiation of CT. In addition, they were advised to have a soft diet. The endpoint was the incidence of OM. Therefore, thorough oral examinations were conducted at each session to identify any areas of OM, and patients were followed for two weeks to evaluate whether LLLT could prevent MTX induced OM.

Treatment of OM

Before any intervention in this randomized clinical trial, patients diagnosed with OM were randomly assigned to either a control group, which received sham laser, or a PDT group, which received the PDT on days 1 and 3 following OM onset. All participants were monitored over a five-day follow-up period. It is important to note that the block randomization with a predefined block size of 4 was performed. Then, sealed envelopes containing the protocol assignments for individual patients were prepared by a member of the academic staff. In the PDT group, ICG (Cardiogreen for microscopy, Sigma-Aldrich, Germany) was used as a photosensitizer and the low-level laser was used as a light source. ICG solution was prepared in distilled water at a concentration of 0.1 mg/ml and applied to the oral lesions using a swab, 10 min prior to laser irradiation. The laser irradiation parameters were identical to those used in the prophylactic LLLT group (laser group), as follows: diode laser (Klas-DX—Intelligent LLLT Dose Unit, Konf, Taiwan) with a continuous wavelength of 808 nm, a spot size of 0.5 cm^2, an energy density of 20 J/cm^2, and an irradiation time of 40 s per point (power: 250 mW).

Clinical assessment of OM

One blinded academic staff member evaluated the patients and selected the cases based on inclusion and exclusion criteria. During each session, a blinded examiner conducted a comprehensive oral examination to assess parameters based on the World Health Organization (WHO) scale, National Cancer Institute (NCI) scale, and Western Consortium for Cancer Nursing Research (WCCNR) stomatitis staging system. The pain and disability were recorded according to the WHO scale: score 0 (none), score 1 (erythema or ulcer without pain), score 2 (painful erythema or ulcer without eating intervention), score 3 (painful erythema or ulcer with eating intervention), score 4 (severe painful lesion needing support). Moreover, the NCI scale was used for erythema, extent, and pseudo membrane formation: score 0 (none), score

1 (mucosal erythema), score 2 (separated pseudo membrane plaques smaller than 1.5 cm), score 3 (confluent pseudo membrane plaques greater than 1.5 cm), and score 4 (tissue necrosis with spontaneous bleeding). It is important to note that the size of lesions was measured by a scaled tongue blade. WCCNR stomatitis staging system also assessed the oral mucosal sites affected by OM based on the number of lesions (0 = no lesion; 1 = 1–4 lesions; 2 = more than four lesions; 3 = coalescing lesion), color (0 = 50% or more pink; 1 = 50% or more slightly red; 2 = 50% or more moderately red; 3 = 50% or more very red), and bleeding (0 = no bleeding; 1 = Not stated; 2 = bleeding with eating or mouth care; 3 = spontaneous bleeding). Thereafter, the scores were summed to compute the final WCCNR score [8].

Statistical analysis

Statistical analysis was conducted using Statistical Package for the Social Sciences (SPSS) software (version 24). The Chi-square test was applied to assess the significance of the difference in the incidence of OM between the groups of the prevention arm. Since the data of subjects in the treatment arm were not normally distributed, nonparametric tests were used: the Friedman test was used for intragroup analysis. Since the *P*-value was less than 0.05, the Wilcoxon test was performed to detect differences in the scales (WHO, NCI, and WCCNR) between the two different sessions (day 1, day 3, and day 5). Moreover, the Mann–Whitney U test was used to compare the reduction in the score of each scale between the PDT group and control group at different follow-up intervals. For all analysis, a *P*-value of less than 0.05 was considered statistically significant.

Results

Patients' characteristics

A total of 45 Iranian patients who met the inclusion and exclusion criteria were selected for the study. Of these, 34 (75.56%) were male and 11 (24.44%) were female, with an age range of 4–17 years (mean 10.9 ± 4.389). A detailed flowchart outlining the progression from participant enrollment to final analysis is presented in Fig. 1.

Outcomes of the preventive protocol

A total of 15 subjects were finally included in the prevention arm and followed for two weeks. From our data, 10 of 15 (66.67%) patients in the control group experienced OM. However, only one patient (6.67%) in the laser group developed OM, while the remaining 14 showed no evidence of OM based on assessments using the WHO, NCI and WCCNR scales. This difference in the incidence of OM between the two groups was statistically significant ($P = 0.001$).

Outcomes of the therapeutic protocol

A total of 30 subjects were finally included in the treatment arm. Descriptive analyses based on the WHO, NCI and WCCNR scales for both control group ($n = 15$) and PDT group ($n = 15$) at different sessions (day 1, day 3, and day 5) are shown in Table 1 and represented graphically in Fig. 2 and Fig. 3. The data indicated that the scores for all scales decreased in both groups over the 5 days of clinical follow-up.

The intragroup analysis of score variations for each scale (WHO, NCI, and WCCNR) between different sessions is shown in Table 2. The Friedman test revealed a significant difference between scores in all scales over the 5 days of clinical follow-up. Since the *P*-value obtained from the Friedman test was less than 0.05, a Post Hoc analysis using the Wilcoxon test was conducted to identify significant differences between the two different sessions. The *P*-values from the both aforementioned tests are presented in Table 2. Moreover, Table 3 lists the comparison of score reductions for each scale between the two groups at different follow-up intervals using the Mann–Whitney test.

Overall, both the PDT group and the control group showed a statistically significant reduction in the score of all scales (WHO, NCI, and WCCNR) from day 1 to day 5 ($P < 0.05$) (Table 2). However, the results of the intergroup analysis, assessed using the Mann–Whitney test, revealed a significant reduction in all scales of the PDT group compared to the control group after 5 days ($P < 0.05$) (Table 3).

In more detail, reductions in WHO scores from day 1 to day 3 were not statistically significant in either group ($P > 0.05$). However, the PDT group demonstrated a statistically significant decrease compared to the control group ($P = 0.041$). In contrast, the WCCNR score reduction in the PDT group from day 1 to day 3 was statistically significant ($P = 0.045$), whereas no significant reduction was observed in the control group ($P > 0.05$). The intergroup analysis using the Mann–Whitney test revealed that this reduction is higher in the PDT group compared to the control group ($P = 0.038$). On the other hand, the WHO and WCCNR scores of both groups showed a significant reduction from day 3 to day 5 ($P < 0.05$). However, intergroup analysis revealed that the differences in these reductions between the two groups were not statistically significant ($P > 0.05$) (Table 2 and Table 3).

Regarding the NCI scale, the Wilcoxon test results also showed that the reductions in the score of this scale from day 1 to day 3 were not statistically significant in either group ($P > 0.05$), but the PDT group demonstrated a statistically significant decrease compared to the control group ($P = 0.033$). The results also showed a statistically significant reduction in the NCI score in the PDT group

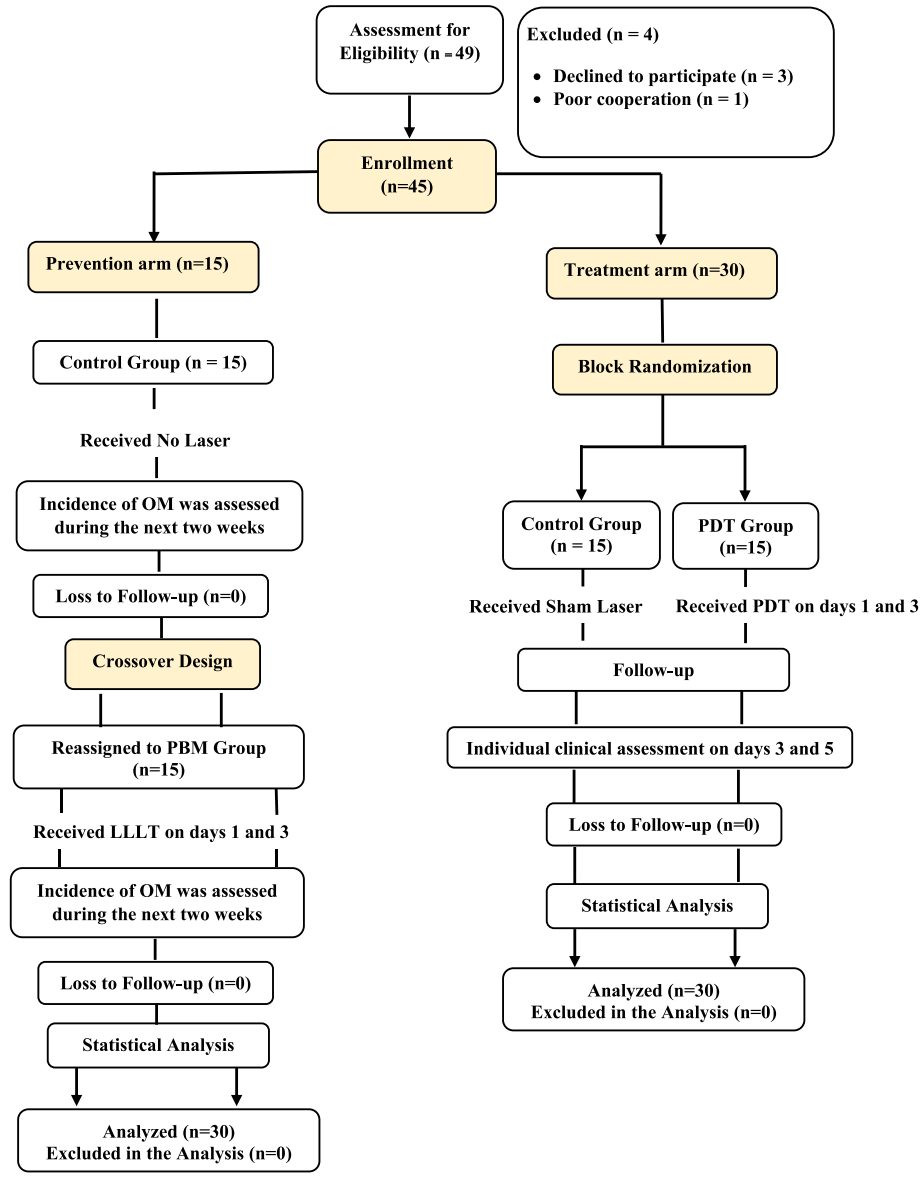


Fig. 1 Study flowchart from enrollment to final analysis

from day 3 to day 5 ($P = 0.003$), whereas no significant reduction was observed in the control group during the same period ($P > 0.05$). The intergroup analysis using the Mann–Whitney test revealed that this reduction is higher in the PDT group compared to the control group ($P = 0.026$) (Table 2 and Table 3).

It is important to note that effect size (ES) values were calculated for all between-group comparisons (Table 3). Notably, all statistically significant comparisons ($p < 0.05$) corresponded with at least medium effect sizes (ES > 0.3), indicating that the study had sufficient power and an adequate sample size to detect moderate to large differences between groups.

Protocol-associated adverse effects

Both preventive and therapeutic protocols were well tolerated, and no adverse effects were reported.

Discussion

MTX, a folic acid antagonist, is an effective drug in the treatment of various pediatric malignancies. In more detail, high doses of MTX are frequently used to treat neoplasms such as ALL and OS. Although it interferes with cell reproduction and demonstrates partial selectivity for tumor cells, it also demonstrates toxicity toward rapidly dividing normal cells, such as those in the gastrointestinal tract [3]. OM is a dose-limiting debilitating

Table 1 The study groups' scales over the 5 days of clinical follow-up

Scale	Control Group		PDT Group	
WHO	**Day 1**		**Day 1**	
	Mean ± SD	2.07 ± 0.704	Mean ± SD	3.2 ± 0.775
	95% CI for mean (LB-UB)	1.68–2.46	95% CI for mean (LB-UB)	2.77—3.63
	Median	2	Median	3
	IQR	2	IQR	2
	Day 3		**Day 3**	
	Mean ± SD	1.93 ± 0.704	Mean ± SD	2.53 ± 0.743
	95% CI for mean (LB-UB)	1.54–2.32	95% CI for mean (LB-UB)	2.12–2.94
	Median	2	Median	3
	IQR	2	IQR	3
	Day 5		**Day 5**	
	Mean ± SD	1.13 ± 0.915	Mean ± SD	1.67 ± 0.724
	95% CI for mean (LB-UB)	0.63–1.64	95% CI for mean (LB-UB)	1.27–2.07
	Median	1	Median	2
	IQR	3	IQR	3
NCI	**Day 1**		**Day 1**	
	Mean ± SD	2.07 ± 0.961	Mean ± SD	3.07 ± 0.799
	95% CI for mean (LB-UB)	1.53–2.60	95% CI for mean (LB-UB)	2.62–3.51
	Median	2	Median	3
	IQR	3	IQR	2
	Day 3		**Day 3**	
	Mean ± SD	2 ± 1.069	Mean ± SD	2.53 ± 0.834
	95% CI for mean (LB-UB)	1.41–2.59	95% CI for mean (LB-UB)	2.07–3.00
	Median	2	Median	2
	IQR	4	IQR	3
	Day 5		**Day 5**	
	Mean ± SD	1.53 ± 1.125	Mean ± SD	1.53 ± 0.743
	95% CI for mean (LB-UB)	0.91–2.16	95% CI for mean (LB-UB)	1.12–1.94
	Median	2	Median	2
	IQR	3	IQR	3
WCCNR	**Day 1**		**Day 1**	
	Mean ± SD	3.8 ± 1.935	Mean ± SD	5.93 ± 2.017
	95% CI for mean (LB-UB)	2.73–4.87	95% CI for mean (LB-UB)	4.82–7.05
	Median	4	Median	5
	IQR	7	IQR	5
	Day 3		**Day 3**	
	Mean ± SD	3.4 ± 1.765	Mean ± SD	4.4 ± 1.595
	95% CI for mean (LB-UB)	2.42–4.38	95% CI for mean (LB-UB)	3.52–5.28
	Median	3	Median	4
	IQR	7	IQR	6
	Day 5		**Day 5**	
	Mean ± SD	2.33 ± 1.799	Mean ± SD	2.87 ± 1.552
	95% CI for mean (LB-UB)	1.34–3.33	95% CI for mean (LB-UB)	2.01–3.73
	Median	2	Median	3
	IQR	7	IQR	6

Abbreviations: CI Confidence Interval, *IQR* Interquartile Range, *LB* Lower Bound, *NCI* National Cancer Institute, *SD* Standard Deviation, *UB* Upper Bound, *WCCNR* Western Consortium for Cancer Nursing Research, *WHO* World Health Organization

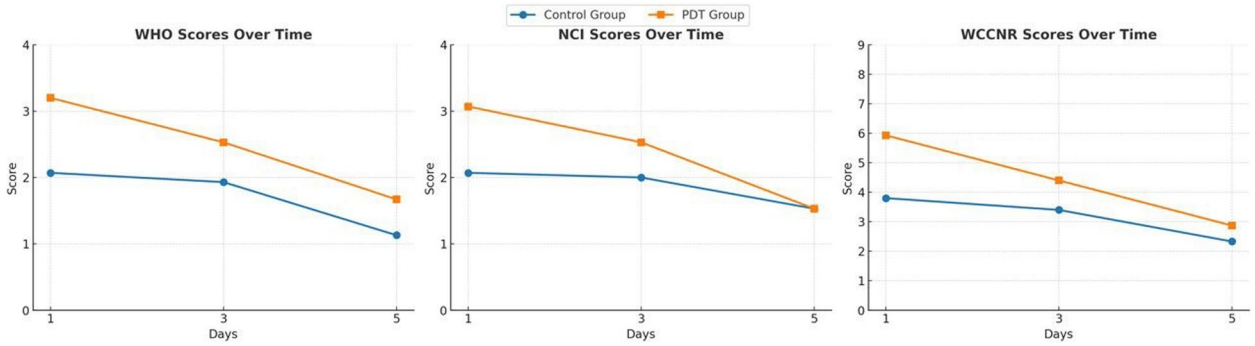


Fig. 2 Trend in mean score changes over time for WHO, NCI, and WCCNR parameters

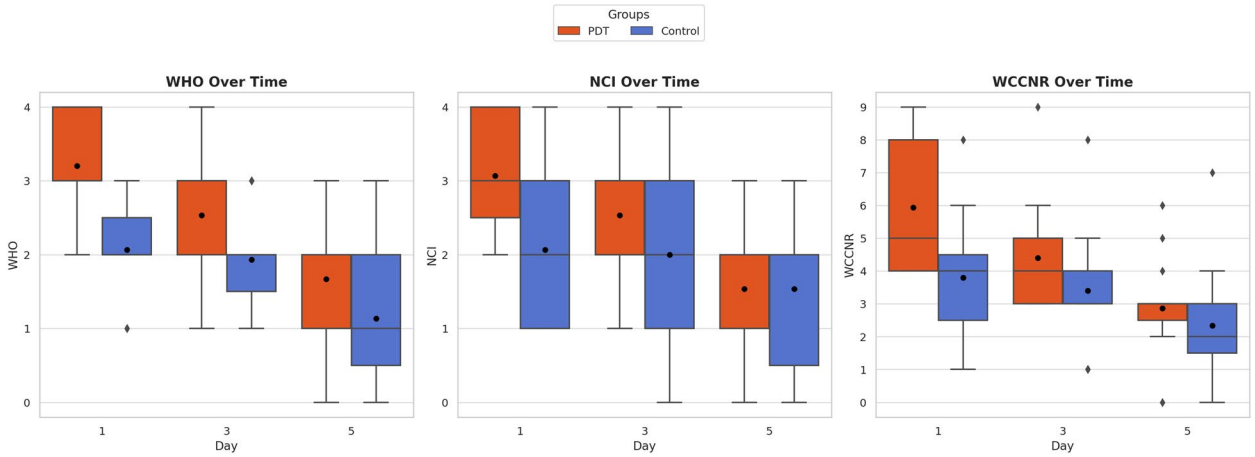


Fig. 3 Box plots showing the distribution of WHO, NCI, and WCCNR scores over time

Table 2 The intragroup analysis (*P*-value) of each scale score variations between different sessions over the 5 days of clinical follow-up

Group	Scale	Friedman Test	Wilcoxon Test		
		Day 1-Day 3- Day 5	Day 1 vs. Day 3	Day 3 vs. Day 5	Day 1 vs. Day 5
Control Group	WHO	0.000*	0.784	0.028*	0.014*
	NCI	0.002*	0.784	0.055	0.028*
	WCCNR	0.000*	0.523	0.002*	0.000*
PDT Group	WHO	0.000*	0.144	0.005*	0.000*
	NCI	0.000*	0.121	0.003*	0.000*
	WCCNR	0.000*	0.045*	0.005*	0.000*

Abbreviations: NCI National Cancer Institute, *WCCNR* Western Consortium for Cancer Nursing Research, *WHO* World Health Organization

* $P < 0.05$ was considered significant

treatment-related complication affecting up to 80% of children receiving CT [11, 12].

OM is characterized by lesions that may be erythematous, atrophic, erosive, or ulcerative. It is mainly caused by two mechanisms: Direct mucosal toxicity and myelosuppression induced by the treatment [11]. In addition to oral lesions, the inflammatory response and disruption of the epithelial barrier in OM result in severe pain and discomfort, impairing the patient´s ability to speak, swallow and eat. Fungal and viral infections may also accompany OM as a result of CT-induced immunosuppression [3]. CT-induced OM typically develops approximately

Table 3 Intergroup comparison of each scale score variations at different follow-up intervals

Scale	Control Group		PDT Group		*P*-Value	ES (r)**
WHO	**Day 3—Day 1**		**Day 3—Day 1**		0.041*	0.373
	Mean ± SD	−0.1333 ± 0.5164	Mean ± SD	−0.6667 ± 0.9759		
	95% CI for mean (LB) – (UB)	(−0.42) – (0.15)	95% CI for mean (LB-UB)	(−1.21) – (−0.13)		
	Median	0	Median	0		
	IQR	2	IQR	3		
	Day 5—Day 3		**Day 5—Day 3**		0.558	0.107
	Mean ± SD	−0.8 ± 0.9411	Mean ± SD	−0.8667 ± 0.6399		
	95% CI for mean (LB-UB)	(−1.32) – (−0.28)	95% CI for mean (LB-UB)	(−1.22) – (−0.51)		
	Median	−1	Median	−1		
	IQR	3	IQR	2		
	Day 5—Day 1		**Day 5—Day 1**		0.047*	0.362
	Mean ± SD	−0.9333 ± 0.9612	Mean ± SD	−1.5333 ± 0.6399		
	95% CI for mean (LB-UB)	(−1.47) – (−0.40)	95% CI for mean (LB-UB)	(−1.89) – (−1.18)		
	Median	−1	Median	−1		
	IQR	3	IQR	2		
NCI	**Day 3—Day 1**		**Day 3 – Day 1**		0.033*	0.389
	Mean ± SD	−0.0667 ± 0.4577	Mean ± SD	−0.5333 ± 0.6399		
	95% CI for mean (LB-UB)	(−0.32) – (0.19)	95% CI for mean (LB-UB)	(−0.89) – (−0.18)		
	Median	0	Median	0		
	IQR	2	IQR	2		
	Day 5—Day 3		**Day 5—Day 3**		0.026*	0.407
	Mean ± SD	−0.4667 ± 0.5164	Mean ± SD	−1 ± 0.6547		
	95% CI for mean (LB-UB)	(−0.75) – (−0.18)	95% CI for mean (LB-UB)	(−1.36) – (−0.63)		
	Median	0	Median	−1		
	IQR	1	IQR	2		
	Day 5—Day 1		**Day 5—Day 1**		0.001*	0.625
	Mean ± SD	−0.5333 ± 0.5164	Mean ± SD	−1.5333 ± 0.9155		
	95% CI for mean (LB-UB)	(−0.82) – (−0.25)	95% CI for mean (LB-UB)	(−2.04) – (−1.03)		
	Median	−1	Median	−1		
	IQR	1	IQR	3		
WCCNR	**Day 3—Day 1**		**Day 3—Day 1**		0.038*	0.378
	Mean ± SD	−0.4 ± 1.1212	Mean ± SD	−1.5333 ± 1.5976		
	95% CI for mean (LB-UB)	(−1.02) – (0.22)	95% CI for mean (LB-UB)	(−2.42) – (−0.65)		
	Median	0	Median	−1		
	IQR	5	IQR	5		
	Day 5—Day 3		**Day 5—Day 3**		0.204	0.232
	Mean ± SD	−1.0667 ± 0.7037	Mean ± SD	−1.5333 ± 1.0601		
	95% CI for mean (LB-UB)	(−1.46) – (−0.68)	95% CI for mean (LB-UB)	(−2.12) – (−0.95)		
	Median	−1	Median	−1		
	IQR	3	IQR	3		
	Day 5—Day 1		**Day 5—Day 1**		0.021*	0.421
	Mean ± SD	−1.4667 ± 1.4075	Mean ± SD	−3.0667 ± 2.0862		
	95% CI for mean (LB-UB)	(−2.25) – (−0.69)	95% CI for mean (LB-UB)	(−4.22) – (−1.91)		
	Median	−1	Median	−3		
	IQR	6	IQR	7		

Abbreviations: CI Confidence Interval, *ES* Effect Size, *IQR* Interquartile Range, *LB*, Lower Bound; NCI, National Cancer Institute; SD, Standard Deviation; UB, Upper Bound; WCCNR, Western Consortium for Cancer Nursing Research; WHO, World Health Organization

* $P < 0.05$ was considered significant

** Effect size was calculated based on the formula: $r = \frac{|Z|}{\sqrt{N}}$

one week after the completion of a CT cycle and varies depending on the dosage and combination of drugs used. Its healing time generally ranges from one to three weeks, depending on the grade of OM [11]. In pediatric patients, severe OM can develop as early as the first week of CT with its incidence increasing over time [2].

LLLT, which operates through non-thermal mechanisms resulting in a temperature increase of less than 1 °C, is a commonly used physiotherapy method that utilizes light energy to influence biological tissues [13, 14]. In fact, light energy is absorbed by cytochromes and porphyrins in the mitochondria. This light activates several cellular pathways, promoting cell proliferation and differentiation, which accelerates regeneration. Additionally, the light triggers pathways that regulate inflammation and reduce pain. The anti-inflammatory, analgesic, and biomodulatory effects of LLLT make it advantageous in treating inflammatory conditions such as OM [14]. While the premise that PBMT (photobiomodulation therapy) effectively modulates various inflammatory processes is certain, the optimal protocol for OM has not yet been established. Key parameters, including wavelength, power, energy density, irradiation time, and continuity, play an essential role in ensuring the safety and effectiveness of the therapy [3, 14, 15]. Underdosing may result in an inadequate cellular response, while overdosing could inhibit cell proliferation or induce apoptosis [5]. Since the early 1980 s, PBMT, formerly known as laser therapy, has been utilized to manage OM. Dr. Ciais and colleagues in France pioneered in its use for cancer patients undergoing combined CT, including 5-FU. They reported a significant reduction in the severity of OM, with oral complications decreasing from 43 to 6% [3]. Although numerous promising prophylactic and therapeutic protocols have been designed to manage OM in patients receiving CT, research on the effectiveness of PBMT, especially in pediatric patients with hematologic cancers, remains limited. To the best of our knowledge, few studies have investigated the efficacy of prophylactic PBM using LLLT in pediatric patients; however, only six have included those with hematologic cancers. Notably, one of these studies (De Castro et al., 2013) focused exclusively on this subgroup [16]. A summary of these studies is presented in Table 4.

Although the results of a study conducted by Cruz et al. in 2007 don't support the prophylactic LLLT when optimal dental and oral care is provided [17], there is adequate agreement among the most previously referenced studies in Table 4 about the benefits of PBM in the prevention of CT induced OM in pediatric patients with hematologic cancers [3, 4, 16, 18, 19]. However, the basic parameters of PBMT reported in these studies varied significantly about types of laser sources (GaAlAs, InGaAlP, and AlGaInAs), wavelengths (660, 670, 685, 780, 830, and 940 nm), and energy densities (4, 4.2, 35, 66.6, and 72 J/cm^2), which makes it difficult to establish the optimal protocol for this age group. In the present study, prophylactic PBMT also led to a statistically significant reduction in the incidence of OM from 66.67% to 6.67% using different laser parameters (wavelength = 808 nm and energy density = 20 J/cm^2) and application schedule (the first irradiation on the day CT starts, followed by an additional session on day 3). The satisfactory outcome of the present study reconfirmed a previous study [3] suggesting prophylactic LLLT in pediatric patients with hematologic cancers (ALL) receiving MTX [3]. Although the prophylactic protocol presented in our study seems to be one of the best available protocols in the literature (Table 4), further studies with larger populations on the ideal parameters of PBMT and application schedule may help improve the outcome.

Oliveira et al. [20] conducted a systematic review and meta-analysis that included data form five articles evaluating the effects of PDT in treating OM. Their findings demonstrated that combining PDT with LLLT is more effective in reducing the healing time of OM lesions compared to LLLT alone. This improved efficacy is due to the combined approach, which addresses both tissue healing and infection control [20]. In fact, PDT is a therapeutic option for managing OM that leverages photophysical and photochemical mechanisms to generate a selective biological response against microorganisms. This approach uses specific light sources combined with photosensitizers such as methylene blue, curcumin, and ICG, demonstrating the potential to treat OM. Among these, methylene blue is the most commonly used photosensitizer, followed by curcumin and ICG [20].

For instance, the efficacy of PDT for the treatment of OM has been demonstrated in a previous study by Lavaee et al. in 2020. They used methylene blue as a photosensitizer and a low-level laser (wavelength = 660 nm) as the light source. While the study did not focus on pediatric patients, it reported significantly lower OM scores (NCI and WCCNR) in the PDT group compared to the control group 3, 5, and 12 days after starting the treatment [8]. Although it is reasonable to hypothesize that interventions effective in adults would yield similar outcomes in children from a biological perspective, this is often not the case. Differences in pharmacokinetics and pharmacodynamics, variations in receptor distribution for specific treatments, and the limited cooperation of pediatric patients are factors that can influence therapeutic outcomes. Therefore, these considerations highlight the importance of conducting additional clinical studies specifically focused on pediatric populations to validate these findings and confirm the efficacy of PDT [21].

Table 4 Studies on prophylactic PBM using LLLT for OM management in pediatric patients, including hematologic cancers

Authors/Year of publication	Participants/ Country	Underlying Conditions	Laser Features			Sites of Low-level Laser Application	Treatment Frequency	Mucositis Scale/ Outcome
			Wave length (nm)	Power (mW)	Fluence (J/cm^2)			
Cruz et al. [17]/2007	**Participants**: Patients aged 3–18 years (Mean ± SD = 8.7 ±4.3), with 39 (65%) being male. Of these, 35 (58%) had leukemia or lymphoma, and 25 (42%) had solid tumors **Country**: Brazil **Sample size**: 60 Control group: 31 Laser group: 29	Patients treated with high-dose MTX (> 1 g/m^2) (23.33%), Conditioning regimen for HSCT (16.67%), ICE regimen (26.67%), Irinotecan (20 mg/m$^{2\times}$ 10) (8.33%), and those with a history of OM from previous CT protocol (25%)	780	60	4	At areas including buccal mucosa, labial mucosa, tongue, soft palate, and sublingual region	5 days following the start of CT, with a median of 4 days (range: 3–5 days)	No significant differences were observed between groups in terms of the prevalence and grade of OM on days 8 and 15 when optimal dental and oral care was provided
Abramoff et al. [4]/2008	**Participants**: Patients aged 7–23 years **Country**: Brazil **Sample size**: 7 patients with a total of 11 CT cycles Control group: 11 Intervention group (LLLT group): 11	Patients undergoing treatment for ALL or OS	685	35	72	At areas including buccal mucosa, labial mucosa, labial commissure, tongue, soft palate, and floor of the mouth	The first irradiation within 24 h of CT initiation, followed by two additional sessions every other day	In the third evaluation, the LLLT group showed significantly lower OM incidence than the control group. According to NCI scale, most patients (73%) in prophylactic LLLT group did not have OM (grade 0) and 18% presented with grade 1. However, only 27% in control group had no OM
De Castro et al. [16]/2013	**Participants**: Patients aged 1–18 years, with 12 (60%) under 8 years All participants (100%) had ALL **Country**: Brazil **Sample size**: 20 Red-subgroup: 10 Infrared-subgroup: 10	Patients with ALL who received High doses of MTX (2.5 and 5.0 g/m2) during the consolidation phase	660 (in red -subgroup) and 830 (in infra-red subgroup)	100	35	At areas including buccal mucosa, labial mucosa, labial commissure, tongue, soft palate, and floor of the mouth	Irradiation at 24-h intervals, starting on Day 1 of the CT cycle and continuing up to Day 5	Prophylactic LLLT showed better outcomes compared to no preventive intervention, with a red laser (660 nm) being more effective than infrared laser (830 nm) in preventing OM

Table 4 (continued)

Authors/Year of publication	Participants/ Country	Underlying Conditions	Laser Features			Sites of Low-level Laser Application	Treatment Frequency	Mucositis Scale/ Outcome
			Wave length (nm)	Power (mW)	Fluence (J/cm^2)			
Ahmed et al. [18]/2015	**Participants**: Pediatric and young adult patients with a mean age of 14.8 years in the control group and 15.6 years in the intervention group. Of the 67 participants, 61.2% were male. Among them, 42 (62.7%) had hematologic cancers, and 25 (37.3%) had solid tumors **Country**: Northern east of Iraq **Sample size**: 67 Control group (sham laser): 33 Intervention group (Active laser): 34	Cancer patients receiving intensive mucotoxic CT, either with a single highly mucotoxic drug or a combination of two or three intensive CT regimens (The most common CT regimen in the sham laser group was MTX-based, while the Etoposide-based regimen was most frequently used in the active laser group)	940 ± 15	0.3	4.2	At areas including buccal mucosa, labial mucosa, tongue, anterior tonsillar pillars, and floor of the mouth	Daily sessions for three consecutive weeks, starting on the first day of the CT course	All the patients were presented with some grade of OM. However, the LLLT group showed significantly lower severity of OM and mean time of healing compared to the control group. Additionally, prophylactic LLLT was associated with a significant reduction in oral pain
Soto et al. [19]/2015	**Participants**: Patients aged 2.1–16.3 years, with a median age of 7.93 in the laser group and 8.63 in the control group. Of the 17 participants, 70.83% were male All participants (100%) received HSCT, and 10 (41.67%) had leukemia or lymphoma **Country**: Brazil **Sample size**: 24 Control group: 12 Intervention group (LLLT group): 12	Oncology patients undergoing HSCT	**Intraoral laser therapy:** 685 **Extraoral laser therapy:** 830	35 80	0.35 J/Point 2.4 J/Point	**Intraoral laser therapy:** At areas including buccal mucosa, labial mucosa, tongue, soft palate, and floor of the mouth **Extraoral laser therapy:** At six points symmetrically distributed across the submandibular and submental areas	4 sessions per week, combining intraoral and extraoral LLLT, with a mean duration of 22 days	The LLLT group showed significantly lower incidence and severity of ulcerative OM and clinical mucositis scores than the control group. Although functional limitations related to diet and swallowing were less severe in the LLLT group, this difference was not statistically significant

Table 4 (continued)

Authors/Year of publication	Participants/ Country	Underlying Conditions	Laser Features			Sites of Low-level Laser Application	Treatment Frequency	Mucositis Scale/ Outcome
			Wave length (nm)	Power (mW)	Fluence (J/cm^2)			
Nunes et al. [3]/2020	**Participants:** Patients aged 8 months to 17 years (Mean ± SD = 9.2 ±4.6), with 82 (55.4%) being male Among them, 74 (50.0%) had ALL, 30 (20.3%) had OS, 23 (15.5%) had AML, six (4.1%) individuals had BL, and 15 (10.1%) were receiving HSCT **Country**: Brazil **Sample size**: 148 • Due to the retrospective design of this study and ethical considerations, a control group of individuals not receiving treatment was not included	Oncology/HSCT patients	660	100	66.6	At areas including lips, buccal mucosa, labial mucosa, tongue, hard and soft palates, and floor of the mouth	Five sessions per week for 1–14 days, before or during CT	LLLT effectively prevented CT-induced OM, particularly in individuals using MTX. The OM grade was 3.16 and 5.45 times higher in individuals with ALL and OS who did not receive prophylactic LLLT compared to those who did. Additionally, prophylactic LLLT was associated with a reduction in episodes of opportunistic infections in ALL patients

Abbreviations: AML Acute Myeloid Leukemia, *ALL* Acute Lymphoblastic Leukemia, *BL* Burkitt Lymphoma, *CT* Chemotherapy, *HSCT* Hematopoietic Stem Cell Transplantation, ICE regimen, Ifosfamide 3 g/m$^{2\times}$ 3, Carboplatin 450 mg/m$^{2\times}$ 2, Etoposide 150 mg/m$^{2\times}$ 3; *MTX* Methotrexate, *NCI* National Cancer Institute *OS* Osteosarcoma

In this trial, we investigated the clinical efficacy of PDT using ICG and low-level laser (808 nm) in the treatment of MTX-induced OM in pediatric patients with hematologic cancers, an area not extensively addressed in previous studies. Although the scores for all scales (WHO, NCI, and WCCNR) decreased in both groups over the 5-day clinical follow-up, the PDT group showed a statistically significant reduction in the scores of all scales at both 3 days and 5 days after starting the treatment compared to the control group. Moreover, the reduction in the NCI score from the third to the fifth days after the starting treatment is higher in the PDT group compared to the control group ($p < 0.05$). Based on the positive results of this study, PDT using ICG and low-level laser is not only safe but also demonstrates significant potential in improving clinical outcomes by alleviating the signs and symptoms of OM in pediatric patients with hematologic cancers. To the best of our knowledge, only two studies to date have focused on PDT for the treatment of OM in pediatric patients, most of whom had hematologic cancers. One is the study by Filho et al. (2017), conducted in Brazil, which demonstrated that the combination of PDT with LLLT enhances therapeutic effects compared to LLLT alone, particularly in reducing lesion size [22]. The other one is the study by da Silva et al. (2018), also conducted in Brazil, which highlighted that PDT could be effectively used for the treatment of OM in children and young patients [21]. Although the photosensitizer and wavelength of the low-level laser used for PDT in these studies differ from those in the present study (methylene blue vs. ICG, and 660nm vs. 808 nm), a consensus is emerging from the findings observed in this study and the literature review that PDT is a safe, practical, and effective treatment option for managing CT-induced OM in pediatric patients, with the potential to improve their quality of life during cancer treatment. Moreover, it can be administered by trained nurses, thereby enhancing the collaborative role of clinical teams.

Although this study provides promising results, its limitations should be considered. First, patients were not followed until the complete resolution of OM after the end of the clinical trial, which may have limited the assessment of the long-term efficacy of PBM. Second, although an initial sample size estimation was performed based on a pilot study, this approach doesn't represent a formal, fully powered sample size calculation. Therefore, the relatively small sample size and dependence on pilot data may limit the statistical power and generalizability of the findings. Nevertheless, we hope that this study will serve as a foundation for future long-term, randomized trials with larger patient populations on this topic to confirm these results.

Conclusion

PBM appears to be a safe and feasible intervention that could be implemented in hematology centers, potentially improving access to this supportive care for MTX-induced OM in pediatric patients with hematologic cancers. In this study, the lower incidence of OM observed in the PBM group, along with encouraging findings from PDT using ICG and low-level laser for OM lesions, highlights the efficacy of PBM for managing MTX-induced OM in pediatric patients with hematologic cancers. However, further studies with larger sample sizes are necessary to confirm these findings and determine the optimal laser parameters and application protocols for this patient population.

Abbreviations

ALL	Acute Lymphoblastic Leukemia
AML	Acute Myeloid Leukemia
BL	Burkitt Lymphoma
CI	Confidence Interval
CT	Chemotherapy
ES	Effect Size
HSCT	Hematopoietic Stem Cell Transplantation
5-FU	5-Fluorouracil
ICE regimen	Ifosfamide 3 g/m^2 × 3, Carboplatin 450 mg/m^2 × 2, Etoposide 150 mg/ m^2 × 3
ICG	Indocyanine green
IQR	Interquartile Range
LB	Lower Bound
LLLT	Low-Level Laser Therapy
MASCC/ISOO	Multinational Association of Supportive Care in Cancer and International Society of Oral Oncology
MTX	Methotrexate
NCI	National Cancer Institute
OM	Oral Mucositis
OS	Osteosarcoma
P	*P*-value
PBM	Photobiomodulation
PBMT	Photobiomodulation therapy
PDT	Photodynamic therapy
SD	Standard Deviation
SPSS	Statistical Package for the Social Sciences
UB	Upper Bound
WCCNR	Western Consortium for Cancer Nursing Research
WHO	World Health Organization

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Authors' contributions

FL was the supervisor and project administrator. FL, FR, and AA constructed the idea for research. FL, FR, and AS planned the methodology. AA evaluated the patients and selected the cases based on inclusion and exclusion criteria. MAA, ND, KF, and AS administered the protocols and evaluated the OM. AS wrote the original draft. FL, FR, and AA reviewed and edited the original draft and all the authors have read and approved the submitted version.

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Data availability
The data supporting the findings of this study are available upon reasonable request from the corresponding author. However, restrictions were applied to the public availability of these data, because of the patients' confidentiality.

Declarations

Ethics approval and consent to participate
The study protocol was according to the ethical committee of the School of Dentistry, Shiraz University of Medical Sciences (Ethical code: IR.SUMS.DENTAL.REC.1401.486). This trial was conducted according to the Declaration of Helsinki. The study was registered on irct.behdasht.gov.ir (Trial Registration Number: IRCT20120101008585 N16). It is important to note that written informed consent was obtained from the legal guardians.

Consent for publication
Not applicable.

Competing of interests
The authors declare no competing interests.

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